

Settings Module – Frontend API

Implementation Guide

Date: 2026-09-09

Base URL path: `/api/v1/settings`

Source of truth: `SettingController` and the Settings request/response DTOs in the backend. Nothing here is invented; anything not defined in the backend is marked **"Not defined in backend"**.

All success responses are wrapped like this (envelope `ApiResponse<T>`):

```
{
  "success": true,
  "message": "Attendance settings updated successfully.",
  "data": { },
  "timestamp": "2026-09-09T14:55:00.123456"
}
```

`data` is the payload documented in §4. All error responses use the format in §10.

1. API List

Method	Endpoint	Purpose	Access Role
GET	<code>/api/v1/settings/attendance</code>	Fetch active attendance settings	SUPER_ADMIN, HR_ADMIN, MANAGER
PUT	<code>/api/v1/settings/attendance</code>	Update attendance settings	SUPER_ADMIN, HR_ADMIN, MANAGER
GET	<code>/api/v1/settings/leave</code>	Fetch active leave settings	SUPER_ADMIN, HR_ADMIN, MANAGER
PUT	<code>/api/v1/settings/leave</code>	Update leave settings	SUPER_ADMIN, HR_ADMIN, MANAGER
GET	<code>/api/v1/settings/company</code>	Fetch active company settings	SUPER_ADMIN, HR_ADMIN, MANAGER
PUT	<code>/api/v1/settings/company</code>	Update company settings	SUPER_ADMIN, HR_ADMIN, MANAGER

There are no other Settings APIs – no create, delete, list, search, pagination, or sorting endpoints exist in the backend.

2. API Details

2.1 Get Attendance Settings

Item	Value
API name	Get Attendance Settings
HTTP method	GET
Endpoint	<code>/api/v1/settings/attendance</code>
Access role	SUPER_ADMIN, HR_ADMIN, MANAGER
Path parameters	None
Query parameters	None
Request body	None
Response body	Envelope with <code>data = AttendanceSettingsResponse</code> (§4.1)
HTTP status codes	200 on success; 404 if the settings row does not exist; 401/403 (see §10)

2.2 Update Attendance Settings

Item	Value
API name	Update Attendance Settings
HTTP method	PUT
Endpoint	<code>/api/v1/settings/attendance</code>
Access role	SUPER_ADMIN, HR_ADMIN, MANAGER
Path parameters	None
Query parameters	None
Request body	<code>AttendanceSettingsRequest</code> (§3.1)

Item	Value
Response body	Envelope with <code>data</code> = <code>AttendanceSettingsResponse</code> (§4.1)
HTTP status codes	200 on success; 400 for malformed body/time format; 404 if the settings row does not exist; 500 if required fields are sent as <code>null</code>

2.3 Get Leave Settings

Item	Value
API name	Get Leave Settings
HTTP method	GET
Endpoint	<code>/api/v1/settings/leave</code>
Access role	SUPER_ADMIN, HR_ADMIN, MANAGER
Path parameters	None
Query parameters	None
Request body	None
Response body	Envelope with <code>data</code> = <code>LeaveSettingsResponse</code> (§4.2)
HTTP status codes	200 on success; 404 if the settings row does not exist; 401/403 (see §10)

2.4 Update Leave Settings

Item	Value
API name	Update Leave Settings
HTTP method	PUT
Endpoint	<code>/api/v1/settings/leave</code>
Access role	SUPER_ADMIN, HR_ADMIN, MANAGER
Path parameters	None
Query parameters	None
Request body	<code>LeaveSettingsRequest</code> (§3.2)
Response body	Envelope with <code>data</code> = <code>LeaveSettingsResponse</code> (§4.2)

Item	Value
HTTP status codes	200 on success; 400 for validation failures; 404 if the settings row does not exist

2.5 Get Company Settings

Item	Value
API name	Get Company Settings
HTTP method	GET
Endpoint	<code>/api/v1/settings/company</code>
Access role	SUPER_ADMIN, HR_ADMIN, MANAGER
Path parameters	None
Query parameters	None
Request body	None
Response body	Envelope with <code>data = CompanySettingsResponse</code> (§4.3)
HTTP status codes	200 on success; 404 (<code>COMPANY_NOT_FOUND</code>) if the settings row does not exist; 401/403 (see §10)

2.6 Update Company Settings

Item	Value
API name	Update Company Settings
HTTP method	PUT
Endpoint	<code>/api/v1/settings/company</code>
Access role	SUPER_ADMIN, HR_ADMIN, MANAGER
Path parameters	None
Query parameters	None
Request body	<code>CompanySettingsRequest</code> (§3.3)

Item	Value
Response body	Envelope with <code>data = CompanySettingsResponse</code> (§4.3)
HTTP status codes	200 on success; 400 for validation failures; 404 if the settings row does not exist; 500 if <code>companyCode</code> violates the unique constraint

3. Request

3.1 AttendanceSettingsRequest (PUT `/api/v1/settings/attendance`)

Field	Data Type	Required	Description
<code>officeStartTime</code>	String (<code>HH:mm:ss</code>)	Yes*	Office start time; after <code>officeStartTime + gracePeriodMinutes</code> , check-in counts as LATE
<code>officeEndTime</code>	String (<code>HH:mm:ss</code>)	Yes*	Office end time; base for early-exit, overtime, missed-checkout cutoff
<code>gracePeriodMinutes</code>	Integer	Yes*	Minutes after office start still treated as on time
<code>minimumWorkingMinutes</code>	Integer	Yes*	Working minutes required to keep PRESENT status at checkout
<code>halfDayWorkingMinutes</code>	Integer	Yes*	Working minutes required for HALF_DAY instead of ABSENT
<code>checkoutCutoffMinutes</code>	Integer	Yes*	Minutes after office end when a missing checkout is auto-marked MISSED_CHECKOUT
<code>overtimeEnabled</code>	Boolean	Yes*	Whether overtime minutes are calculated
<code>weekendAttendanceAllowed</code>	Boolean	Yes*	Whether check-in is allowed on Saturday/Sunday
<code>holidayAttendanceAllowed</code>	Boolean	Yes*	Whether check-in is allowed on holidays that disallow attendance

*

JSON request example:

```
{
  "officeStartTime": "09:30:00",
  "officeEndTime": "18:30:00",
  "gracePeriodMinutes": 45,
  "minimumWorkingMinutes": 480,
  "halfDayWorkingMinutes": 240,
  "checkoutCutoffMinutes": 180,
  "overtimeEnabled": false,
  "weekendAttendanceAllowed": false,
  "holidayAttendanceAllowed": false
}
```

3.2 LeaveSettingsRequest (PUT </api/v1/settings/leave>)

Field	Data Type	Required	Description
<code>carryForwardAllowed</code>	Boolean	Yes (<code>@NotNull</code>)	<code>true</code> = unused monthly leave carries into the annual balance; <code>false</code> = expires at month-end
<code>monthlyGuideline</code>	Integer	Yes (<code>@NotNull</code> , <code>@Min(0)</code>)	Recommended paid-leave days per month; also used by payroll as allowed monthly paid leave
<code>annualPaidLeave</code>	Integer	Yes (<code>@NotNull</code> , <code>@Min(0)</code>)	Total annual paid-leave days; used as fallback allocation for paid leave types

JSON request example:

```
{
  "carryForwardAllowed": false,
  "monthlyGuideline": 2,
  "annualPaidLeave": 24
}
```

3.3 CompanySettingsRequest (PUT </api/v1/settings/company>)

Field	Data Type	Required	Description
<code>companyName</code>	String (≤150)	Yes (<code>@NotBlank</code>)	Company display name

Field	Data Type	Required	Description
<code>companyCode</code>	String (≤30)	Yes (@NotBlank)	Short company code – unique in the database
<code>email</code>	String (≤150)	Yes (@NotBlank , @Email)	Company contact email
<code>phoneNumber</code>	String (≤20)	No	Digits only, 10–15 characters (<code>^[0-9]{10,15}\$</code>)
<code>website</code>	String (≤150)	No	Company website
<code>addressLine1</code>	String (≤255)	No	Address line 1
<code>addressLine2</code>	String (≤255)	No	Address line 2
<code>city</code>	String (≤100)	No	City
<code>state</code>	String (≤100)	No	State
<code>country</code>	String (≤100)	No	Country
<code>postalCode</code>	String (≤20)	No	Postal code
<code>timeZone</code>	String (≤50)	Yes (@NotBlank)	Company time zone (informational display)
<code>currency</code>	String (≤10)	Yes (@NotBlank)	Company currency
<code>workingDaysPerWeek</code>	Integer	No (@Min (1) , @Max (7))	Working days per week (1–7)

Note: `logoUrl` is **not part of the request** – it cannot be changed through the API.

JSON request example:

```
{
  "companyName": "MyHourly",
  "companyCode": "MHR",
  "email": "admin@myhourly.com",
  "phoneNumber": "9876543210",
  "website": "https://myhourly.com",
```

```
"addressLine1": "Plot 12, Tech Park",
"addressLine2": "HITEC City",
"city": "Hyderabad",
"state": "Telangana",
"country": "India",
"postalCode": "500081",
"timeZone": "Asia/Kolkata",
"currency": "INR",
"workingDaysPerWeek": 5
}
```

4. Response

All responses use the envelope from the top of this document; the tables below describe `data`.

4.1 AttendanceSettingsResponse

Field	Data Type	Nullable	Description
<code>id</code>	Long	No	Settings row ID
<code>officeStartTime</code>	String (<code>HH:mm:ss</code>)	No	Office start time
<code>officeEndTime</code>	String (<code>HH:mm:ss</code>)	No	Office end time
<code>gracePeriodMinutes</code>	Integer	No	Grace period in minutes
<code>minimumWorkingMinutes</code>	Integer	No	Full-day working-minute threshold
<code>halfDayWorkingMinutes</code>	Integer	No	Half-day working-minute threshold
<code>checkoutCutoffMinutes</code>	Integer	No	Missed-checkout cutoff in minutes after office end
<code>overtimeEnabled</code>	Boolean	No	Whether overtime is calculated
<code>weekendAttendanceAllowed</code>	Boolean	No	Whether weekend check-in is allowed
<code>holidayAttendanceAllowed</code>	Boolean	No	Whether holiday check-in is allowed
<code>active</code>	Boolean	No	Always <code>true</code> in practice; no API to change it

JSON response example (the `data` object):

```
{
  "id": 1,
  "officeStartTime": "09:30:00",
  "officeEndTime": "18:30:00",
  "gracePeriodMinutes": 45,
  "minimumWorkingMinutes": 480,
  "halfDayWorkingMinutes": 240,
  "checkoutCutoffMinutes": 180,
  "overtimeEnabled": false,
  "weekendAttendanceAllowed": false,
  "holidayAttendanceAllowed": false,
  "active": true
}
```

Success message values: "Attendance settings fetched successfully." (GET),
"Attendance settings updated successfully." (PUT).

4.2 LeaveSettingsResponse

Field	Data Type	Nullable	Description
<code>id</code>	Long	No	Settings row ID
<code>carryForwardAllowed</code>	Boolean	No	Whether unused monthly leave carries forward
<code>monthlyGuideline</code>	Integer	No	Recommended monthly paid-leave days
<code>annualPaidLeave</code>	Integer	No	Annual paid-leave days (fallback allocation)
<code>active</code>	Boolean	No	Always <code>true</code> in practice

JSON response example (the `data` object):

```
{
  "id": 1,
  "carryForwardAllowed": false,
  "monthlyGuideline": 2,
  "annualPaidLeave": 24,
  "active": true
}
```

Success message values: "Leave settings fetched successfully." (GET), "Leave settings updated successfully." (PUT).

4.3 CompanySettingsResponse

Field	Data Type	Nullable	Description
<code>id</code>	Long	No	Settings row ID
<code>companyName</code>	String	No	Company display name
<code>companyCode</code>	String	No	Short company code (unique)
<code>email</code>	String	No	Company contact email
<code>phoneNumber</code>	String	Yes	Company phone
<code>website</code>	String	Yes	Company website
<code>addressLine1</code>	String	Yes	Address line 1
<code>addressLine2</code>	String	Yes	Address line 2
<code>city</code>	String	Yes	City
<code>state</code>	String	Yes	State
<code>country</code>	String	Yes	Country
<code>postalCode</code>	String	Yes	Postal code
<code>timeZone</code>	String	No	Company time zone
<code>currency</code>	String	No	Company currency
<code>workingDaysPerWeek</code>	Integer	No	Working days per week (1-7)
<code>logoUrl</code>	String	Yes	Logo URL – read-only ; not updatable via the API
<code>active</code>	Boolean	No	Always <code>true</code> in practice

JSON response example (the `data` object):

```
{
  "id": 1,
  "companyName": "MyHourly",
  "companyCode": "MHR",
  "email": "admin@myhourly.com",
  "phoneNumber": "9876543210",
  "website": null,
  "addressLine1": null,
```

```
"addressLine2": null,
"city": null,
"state": null,
"country": null,
"postalCode": null,
"timeZone": "Asia/Kolkata",
"currency": "INR",
"workingDaysPerWeek": 5,
"logoUrl": null,
"active": true
}
```

Success message values: "Company settings fetched successfully." (GET), "Company settings updated successfully." (PUT).

(The timestamp field of `createdAt` / `updatedAt` audit columns is not exposed in any Settings response – only `id` and `active` from the base class are mapped.)

5. Enums

Settings request/response DTOs contain **no enum fields**. The only enum that can appear in a Settings API exchange is the error code in error responses:

Enum	Values	Used In
<code>ErrorCode</code> (in error payloads)	Relevant to Settings: <code>VALIDATION_FAILED</code> , <code>INVALID_REQUEST</code> , <code>RESOURCE_NOT_FOUND</code> , <code>COMPANY_NOT_FOUND</code> , <code>ACCESS_DENIED</code> , <code>INTERNAL_SERVER_ERROR</code>	<code>errorCode</code> field of every error response (§10)

(The backend `ErrorCode` enum contains many more values used by other modules; only the ones above can be returned by Settings APIs.)

6. Access Roles

All six Settings endpoints share the same role set:

API	Role/Permission	Access
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API	Role/Permission	Access
GET <code>/api/v1/settings/attendance</code>	<code>SUPER_ADMIN</code> , <code>HR_ADMIN</code> , <code>MANAGER</code>	Allowed for all three; denied for all other roles
PUT <code>/api/v1/settings/attendance</code>	<code>SUPER_ADMIN</code> , <code>HR_ADMIN</code> , <code>MANAGER</code>	Same as above
GET <code>/api/v1/settings/leave</code>	<code>SUPER_ADMIN</code> , <code>HR_ADMIN</code> , <code>MANAGER</code>	Same as above
PUT <code>/api/v1/settings/leave</code>	<code>SUPER_ADMIN</code> , <code>HR_ADMIN</code> , <code>MANAGER</code>	Same as above
GET <code>/api/v1/settings/company</code>	<code>SUPER_ADMIN</code> , <code>HR_ADMIN</code> , <code>MANAGER</code>	Same as above
PUT <code>/api/v1/settings/company</code>	<code>SUPER_ADMIN</code> , <code>HR_ADMIN</code> , <code>MANAGER</code>	Same as above

7. Datatypes

Data Type	JSON appearance	Settings fields using it
Long (number)	<code>1</code>	<code>id</code> in all responses
Integer (number)	<code>45</code>	<code>gracePeriodMinutes</code> , <code>minimumWorkingMinutes</code> , <code>halfDayWorkingMinutes</code> , <code>checkoutCutoffMinutes</code> , <code>monthlyGuideline</code> , <code>annualPaidLeave</code> , <code>workingDaysPerWeek</code>
Boolean	<code>true</code> / <code>false</code>	<code>overtimeEnabled</code> , <code>weekendAttendanceAllowed</code> , <code>holidayAttendanceAllowed</code> , <code>carryForwardAllowed</code> , <code>active</code>
String (time)	<code>"09:30:00"</code>	<code>officeStartTime</code> , <code>officeEndTime</code>
String	<code>"MyHourly"</code>	All company fields
Date/DateTime	<code>"2026-09-09T14:55:00.123456"</code>	<code>timestamp</code> in the response envelope

Data Type	JSON appearance	Settings fields using it
Enum	"VALIDATION_FAILED"	<code>errorCode</code> in error responses only
Double / BigDecimal / UUID / List / Object	—	Not used in any Settings API

Date/time formats defined by the backend:

Type	Format	Notes
Time (<code>LocalTime</code>)	HH:mm:ss	Malformed values return 400 with message "Invalid time format. Expected format: HH:mm:ss (example: 18:30:00)"
DateTime (<code>LocalDateTime</code>)	yyyy-MM-dd'T'HH:mm:ss (ISO, e.g. 2026-09-09T14:55:00.123456)	Used for the envelope <code>timestamp</code>
LocalDate	yyyy-MM-dd	Not used by Settings (global error-handler format listed for completeness)

8. Validation

Attendance (PUT `/attendance`) – backend validates nothing

Field	Required	Validation
all 9 fields	Effectively yes (see §3.1 warning)	Not defined in backend – the frontend must enforce: time format <code>HH:mm:ss</code> , non-negative minutes, sensible ranges. Sending <code>null</code> for any field causes HTTP 500 (DB <code>NOT NULL</code>).

Leave (PUT `/leave`) – validated by backend

Field	Required	Validation
<code>carryForwardAllowed</code>	Yes	<code>@NotNull</code>
<code>monthlyGuideline</code>	Yes	<code>@NotNull</code> , <code>@Min(0)</code>
<code>annualPaidLeave</code>	Yes	<code>@NotNull</code> , <code>@Min(0)</code>

Company (PUT /company) – validated by backend

Field	Required	Validation
companyName	Yes	@NotBlank ("Company name is required."), @Size(max = 150)
companyCode	Yes	@NotBlank ("Company code is required."), @Size(max = 30) ; unique in DB (no friendly duplicate error – see §10)
email	Yes	@NotBlank ("Email is required."), @Email ("Invalid email format.")
phoneNumber	No	@Pattern(regex = "[0-9]{10,15}\$") ("Phone number must contain 10 to 15 digits.")
website	No	@Size(max = 150)
addressLine1	No	@Size(max = 255)
addressLine2	No	@Size(max = 255)
city	No	@Size(max = 100)
state	No	@Size(max = 100)
country	No	@Size(max = 100)
postalCode	No	@Size(max = 20)
timeZone	Yes	@NotBlank ("Time zone is required.")
currency	Yes	@NotBlank ("Currency is required.")
workingDaysPerWeek	No	@Min(1) , @Max(7)

Behavior on validation failure: the backend returns **only the first** failing message (HTTP 400). Run complete frontend validation first so users can fix everything in one pass.

9. Default Values

Default values expected on a fresh environment:

Setting	Data Type	Default Value
officeStartTime	String (time)	"09:30:00"

Setting	Data Type	Default Value
<code>officeEndTime</code>	String (time)	<code>"18:30:00"</code>
<code>gracePeriodMinutes</code>	Integer	<code>45</code>
<code>minimumWorkingMinutes</code>	Integer	<code>480</code>
<code>halfDayWorkingMinutes</code>	Integer	<code>240</code>
<code>checkoutCutoffMinutes</code>	Integer	<code>180</code>
<code>overtimeEnabled</code>	Boolean	<code>false</code>
<code>weekendAttendanceAllowed</code>	Boolean	<code>false</code>
<code>holidayAttendanceAllowed</code>	Boolean	<code>false</code>
<code>carryForwardAllowed</code>	Boolean	<code>false</code>
<code>monthlyGuideline</code>	Integer	<code>2</code>
<code>annualPaidLeave</code>	Integer	<code>24</code>
<code>companyName</code>	String	<code>"MyHourly"</code>
<code>companyCode</code>	String	<code>"MHR"</code>
<code>email</code> (company)	String	Value of the backend property <code>app.super-admin.email</code>
<code>phoneNumber</code>	String	<code>"9876543210"</code>
<code>timeZone</code>	String	<code>"Asia/Kolkata"</code>
<code>currency</code>	String	<code>"INR"</code>
<code>workingDaysPerWeek</code>	Integer	<code>5</code>
<code>active</code> (all settings rows)	Boolean	<code>true</code>

The following fields have no default value: `website` , `addressLine1` , `addressLine2` , `city` , `state` , `country` , `postalCode` , `logoUrl` (all stay `null`). There is **no "reset to defaults" API** — the table is for reference/placeholders only.

10. Error Response

Every error (all Settings endpoints) uses this shape (`ApiError`):

```
{
  "success": false,
  "message": "Time zone is required.",
  "errorCode": "VALIDATION_FAILED",
  "path": "/api/v1/settings/company",
  "timestamp": "2026-09-09T14:55:00.123456"
}
```

HTTP Status	When it happens	errorCode	Example message
400	Bean-Validation failure on leave/company PUT (@NotNull , @NotBlank , @Email , @Min , @Size , @Pattern)	VALIDATION_FAILED	"Company code is required." – only the first failing field's message is returned
400	Service-level rule violation	VALIDATION_FAILED	Currently none active for Settings – leave settings validation is commented out in the backend
400	Malformed JSON body, wrong time format, wrong datatype	INVALID_REQUEST	"Invalid time format. Expected format: HH:mm:ss (example: 18:30:00)" / "Invalid request format"
401	Missing/expired authentication token	(security-layer response)	Handle globally in the app, same as other APIs
403	Authenticated user without SUPER_ADMIN / HR_ADMIN / MANAGER role	ACCESS_DENIED	"Access denied."
404	Settings row does not exist (initialization never ran / row missing)	RESOURCE_NOT_FOUND (attendance, leave)	"Attendance settings not found." / "Leave settings not found."
404	Company settings row does not exist	COMPANY_NOT_FOUND	"Company settings not found."
500	Unexpected backend error – including a duplicate <code>companyCode</code> violating the DB unique constraint on PUT <code>/company</code>	INTERNAL_SERVER_ERROR	"An unexpected error occurred. (...)"

Frontend handling checklist per call: show `message` from the error body in a toast/inline alert; on 403 hide the form; on 404 show a "Settings not initialized" empty state; on 500 show a generic error (for

company updates, warn the user if they changed `companyCode`).

Final API Reference

API	Method	Endpoint	Role	Request	Response
Get Attendance Settings	GET	<code>/api/v1/settings/attendance</code>	SUPER_ADMIN, HR_ADMIN, MANAGER	—	<code>AttendanceSettingsResponse</code>
Update Attendance Settings	PUT	<code>/api/v1/settings/attendance</code>	SUPER_ADMIN, HR_ADMIN, MANAGER	<code>AttendanceSettingsRequest</code>	<code>AttendanceSettingsResponse</code>
Get Leave Settings	GET	<code>/api/v1/settings/leave</code>	SUPER_ADMIN, HR_ADMIN, MANAGER	—	<code>LeaveSettingsResponse</code>
Update Leave Settings	PUT	<code>/api/v1/settings/leave</code>	SUPER_ADMIN, HR_ADMIN, MANAGER	<code>LeaveSettingsRequest</code>	<code>LeaveSettingsResponse</code>
Get Company Settings	GET	<code>/api/v1/settings/company</code>	SUPER_ADMIN, HR_ADMIN, MANAGER	—	<code>CompanySettingsResponse</code>
Update Company Settings	PUT	<code>/api/v1/settings/company</code>	SUPER_ADMIN, HR_ADMIN, MANAGER	<code>CompanySettingsRequest</code>	<code>CompanySettingsResponse</code>